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Intervention

A cognitive behavior therapy-based intervention among poorly controlled adult type 1 diabetes patients—A randomized controlled trial

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ABSTRACT

Objective: To examine the impact of a Cognitive Behavior Therapy (CBT)-based intervention on HbA_{1c}, self-care behaviors and psychosocial factors among poorly controlled adult type 1 diabetes patients.
Methods: Ninety-four type 1 diabetes patients were randomly assigned to either an intervention group or a control group. The intervention was based on CBT and was mainly delivered in group format, but individual sessions were also included. All subjects were provided with a continuous glucose monitoring system (CGMS) during two 3-day periods. HbA_{1c}, self-care behaviors and psychosocial factors were measured up to 48 weeks.
Results: Significant differences were observed with respect to HbA_{1c} ($P < 0.05$), well-being ($P < 0.05$), diabetes-related distress ($P < 0.01$), frequency of blood glucose testing ($P < 0.05$), avoidance of hypoglycemia ($P < 0.01$), perceived stress ($P < 0.05$), anxiety ($P < 0.05$) and depression ($P < 0.05$), all of which showed greater improvement in the intervention group compared with the control group. A significant difference ($P < 0.05$) was registered with respect to non-severe hypoglycemia, which yielded a higher score in the intervention group.
Conclusion: This CBT-based intervention appears to be a promising approach to diabetes self-management.
Practice implications: Diabetes care may benefit from applying tools commonly used in CBT. For further scientific evaluation in clinical practice, there is a need for specially educated diabetes care teams, trained in the current approach, as well as cooperation between diabetes care teams and psychologists trained in CBT.

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1. Introduction

Concrete everyday-life behavior is an essential part of diabetes management in order to control blood glucose. The daily demands of diabetes are well documented [1,2], as are the beneficial effects of good glycemic control on long-term complications [3]. In clinical practice, however, many patients remain in poor glycemic control and have problems adhering to the treatment [4]. Barriers to optimal self-care and glycemic control may be numerous and may vary from an individual perspective. The management of type 1 diabetes in itself presents numerous challenges, including many that must be addressed daily and may be burdensome, such as self-injections of insulin and frequent monitoring of blood glucose [5]. There is also a risk of type 1 diabetes patients developing fear of hypoglycemia, leading them to underdose their insulin and to overcorrect low blood glucose levels, resulting in hyperglycemia [6,7]. To enhance patients' self-management and well-being, an overall behavior-oriented approach to diabetes management has been proposed [8–11], including steps such as constructing a problem definition, collaborative goal-setting, collaborative problem-solving, contract for change and continuing support [8]. The present intervention approach was based on Cognitive Behavior Therapy (CBT), in which a basic premise is the existence of a constant interplay between the internal and external environment affecting the individual's overall functioning. Thus, overt behavior (actions), cognitions (thoughts, beliefs, perceptions), feelings, and physiology are tightly and interactively integrated, and will therefore affect the way the patient manages his/her diabetes as

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a whole [12–14]. Based on this, it was assumed that a CBT-based intervention could raise participants' awareness of these processes and help them to change concrete self-care behaviors by using specific, targeted tools [12,13], thus enabling them to better cope with their diabetes. It was hypothesized that this intervention would improve both relevant psychological factors and self-care behaviors, resulting in better glycemic control.

Interventions based on cognitive behavior therapy have previously been successfully applied in health behavior change in different areas [15–19]. In the field of diabetes, a broad spectrum of behavior-oriented interventions has shown promising results on various psychological variables, but has not generated such convincing long-term effects regarding improvements in glycemic control [20–23]. Reviews, to date, indicate a lack of solid evidence regarding such interventions, particularly for those adult type 1 diabetes patients who are poorly controlled [9,10,24]. A recent meta-analysis [25] found evidence for the effectiveness of psychological treatments in improving glycemic control in children and adolescents, but not in adults with type 1 diabetes. To our knowledge, the only randomized controlled trial to test a CBT-based program on this targeted group of patients is the study by van der Ven et al. [23]. They found that their program was successful in improving self-efficacy, diabetes-related distress and mood at 3-month follow-up, but found no improvements in glycemic control.

A closer look at the studies above, and in line with some of the suggestions made by the authors themselves, reveals possible limitations. For example, some of the programs are too short and not sufficiently tailored to the patients' individual needs [23], there is a lack of a structured maintenance program components for the behavior changes achieved and there may not have been enough focus on specific behavioral components [22]. Therefore, in the current study we chose to include such apparently missing intervention components, thus incorporating a combined individual and group format, a structured maintenance program for behavior change and biofeedback, focusing on analyzing the glycemic effects of actual self-care behaviors. Here, we report on the effects after 48 weeks, of this structured and expanded CBT-based intervention. The aim of the study was to examine the impact of the intervention on HbA_{1c}, self-care behaviors and psychosocial factors in poorly controlled adult type 1 diabetes patients.

2. Methods

2.1. Study design

This randomized controlled trial was preceded by an uncontrolled pilot test in six patients, supporting its feasibility.

A total of 32 patients were required in each group in order to achieve 80% power to detect an absolute difference between groups at week 48, in $HbA1_c$ levels of 0.5% at the 5% level of significance, assuming an SD of 0.7%. We recruited 47 patients in each group to allow for attrition. The choice of effect size for power analyses was based on data from the DCCT [3] with a 0.5% difference regarded as being clinically significant. This was also supported by the procedure reported by van der Ven et al. [23].

After approval by the local Ethics Committee, 94 participants were consecutively recruited from a sample of 230 eligible patients, meeting study criteria of a clinical diagnosis of type 1 diabetes, duration of at least two years, age 18–65 years, BMI < 30 kg/m^2, $HbA_{1c} > 7.5\%$ (reference $< 5.2\%$) during the last year. Inclusion and exclusion criteria were retrieved from the patients' medical records and confirmed by means of a telephone interview carried out by the diabetes specialist nurse. All patients had received a basic diabetes education program and attended one of the two diabetes out-patient clinics regularly. Reasons for excluding patients from the study were: insufficient reading and comprehension skills, pregnancy, diagnosed psychiatric illness or alcohol and/or drug abuse, and an ongoing intercurrent disease. Recruitment took place at the diabetes care units of two university hospitals in Stockholm during the years 2005 and 2006.

Patients who fulfilled the inclusion criteria were informed by regular mail, and one week later they were contacted by phone and invited to participate. A total of 122 (53%) patients declined. Those who gave their informed consent to participate received a battery of self-reported questionnaires together with a pre-paid envelope. When all questionnaires had been filled out by the patients and returned to the research clinic, a sex-stratified randomization was executed manually by a person not involved in the study. The intention was to get mixed groups of females and males. The randomization was done in blocks of 16 patients, eight in each arm. The flow diagram (Fig. 1) shows the trial profile.

2.2. The CBT-based intervention

Patients randomized to the intervention group first had a meeting with the diabetes specialist nurse. At this time point, background characteristics were determined, as well as self-care behaviors and current problem areas in diabetes.

The intervention entitled StyrKRAFT i Ditt Liv©, meaning "Power to Choose your Direction", was designed by the research group and was based on CBT principles. In part, the model applied principles, practical techniques and first-hand experience from other comparable areas of intervention [17]. The intervention was led by a diabetes specialist nurse (first author) and a psychologist trained in CBT (second author), and was delivered in closed groups of 4–6 patients.

A structured manual, incorporating all the sessions, was developed. This consisted of a *basic intervention* program (weeks 0–8), where the main purpose was to map the patients' own behaviors using a logbook of daily self-register, and teach them different tools suitable for working towards behavior change, as well as a *structured maintenance* program (weeks 9–48), where the main focus was on maintaining the behavior changes achieved and on tackling future risk of relapses.

2.2.1. Basic intervention program

The basic intervention program consisted of 8 weekly 2-hour sessions: with the exception of session 7, all were delivered in groups. These basic sessions, as presented in Table 1, followed the standard format of an initial and brief relaxation training period, following the "Applied Tension Release, ATR" format [26], review of homework with focus on self-care, introduction of a new theme and a related tool suitable for behavior modification [12,27–29]. The detailed outline of sessions was developed by the psychologist in conjunction with the diabetes specialist nurse and inspiration was derived from clinical experience with diabetes patients and from previous research in the area [20,30,31]. On two occasions the patients were provided with a continuous glucose monitoring system (CGMS Gold, Medtronic Minimed, Northridge, CA, USA) and received instructions on its use over 72 h. This CGMS procedure was repeated two weeks later, and the data obtained herewith served as biofeedback [32], supported by the diabetes specialist nurse. The main purpose of this pedagogic strategy was not to educate the patient in general, but rather to help him/her reflect upon and analyze current self-care behaviors, and especially how these relate to the momentary glucose profiles.

The patients were assigned weekly homework to be done individually. Both written and oral feedback of homework was given in order to encourage further reflections and actions among participants. During the seventh individual session, held by either

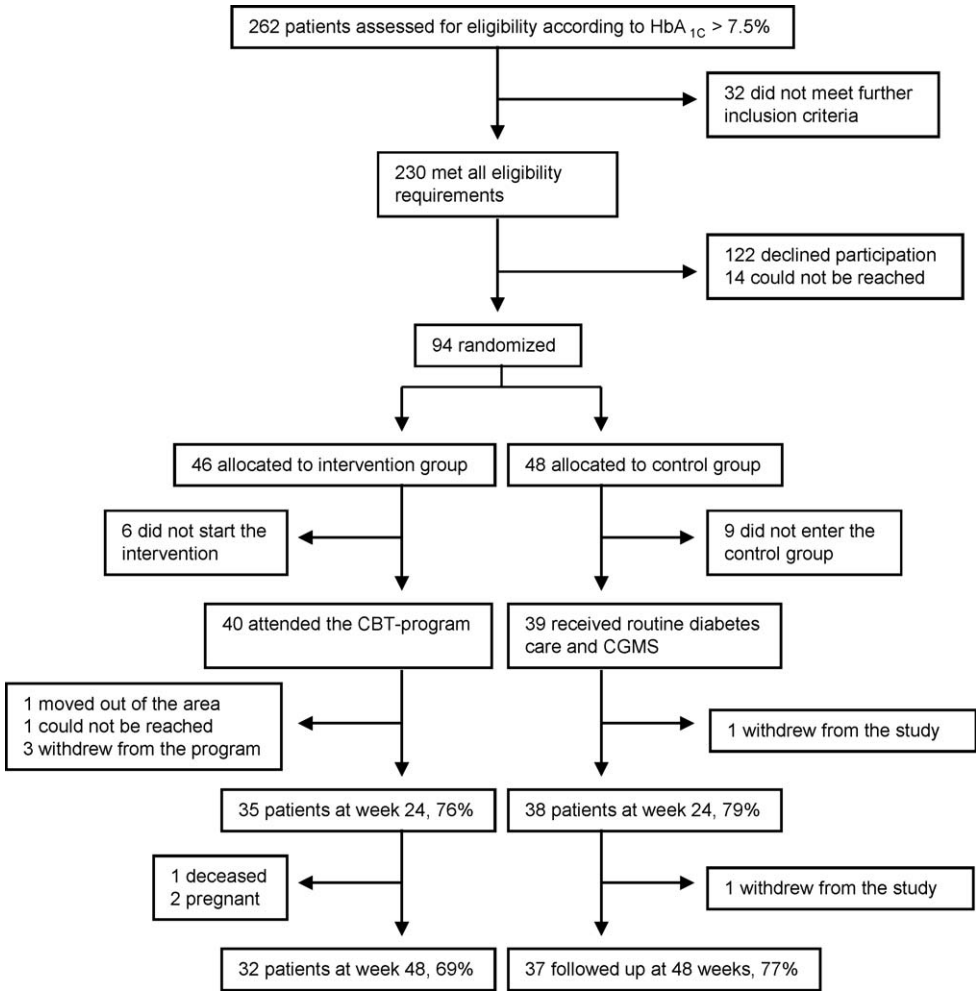

Fig. 1. Flow of participants through the study.

of the therapists, each participant designed a plan for his/her behavior-change activities.

Parallel with the CBT-based intervention, all patients continued receiving traditional diabetes care and were free to modify their insulin regimen as required.

2.2.2. Maintenance program

During the structured maintenance program the plan for behavior change was followed up in two group sessions, as well as two individual sessions at weeks 12 and 24. Five phone calls from either the psychologist or the diabetes specialist nurse were also included during weeks 10–42. Each participant had regular contact with one and the same therapist during the maintenance phase, which included feedback on behavioral changes as well as problem-solving and replanning discussions, as required.

2.3. Control group

A research assistant performed all initial assessments and routines regarding CGMS in the control group in order to control for contamination bias. The patients were not given any structured feedback on the CGMS, but were free to discuss the glucose profiles with their diabetes specialist nurse or the diabetes physician at their regular appointments. Apart from CGMS, the patients continued receiving routine diabetes care and were put on a waiting list to take part in the CBT-based intervention if the results of the analysis after one year proved to be promising.

2.4. Measures

The primary outcome variable was glycemic control (HbA_{1c}). Secondary variables were a set of variables from self-reported questionnaires, chosen on the basis of the contents and the central themes of the intervention, and also with regard to known challenging areas for type 1 diabetes patients related to metabolic control [7,31,33–36].

2.4.1. Glycemic control

HbA_{1c} was analyzed with filter paper technique (HbA_{1c} via post) at Karolinska University Laboratory, using an immunological assay developed by Roche [37]. Time points for measurement were: at baseline, and after 8, 16, 24, 32, 40 and 48 weeks (Table 1).

2.4.2. Self-care behaviors

Frequency of self-care behaviors over the past week was assessed with the revised version of the 17-item Summary of Self-Care Activities (SDSCA) measure [38]. In this study we used the four two-item subscales regarding general diet, specific diet, exercise and blood sugar testing. With respect to our study sample of type 1 diabetes patients, two additional items were used regarding medication and hypoglycemia from the Diabetes Self-Care Inventory (DSCI) [30]. Due to the multidimensional nature of self-care, it is necessary to assess each component separately rather than produce a total score of the measure [38]. Higher scores indicate higher levels of self-care. Internal consistency, measured

Table 1
Overview of the CBT-based intervention.

Week	Session	Theme	Tools and homework issues	Measure
0				Δ
Basic program				
1	1 (G)	How does diabetes affect my life?	Log book for self-care activities and feelings of stress. Interaction model (the interplay between the internal and external environment). Applied Tension Release (long version).	
2	2 (G)	Stress and diabetes. CGMS.	Log book. CGMS. Negative automatic thoughts—how to identify them. Problem-solving.	
3	3 (G)	Stress and diabetes, continued. Feedback on CGMS.	Log book. Negative automatic thoughts—how to handle them. Applied Tension Release (short version).	
4	4 (G)	Diabetic complications and the future.	Log book. Coping with worries/anxiety. Exposure.	
5	5 (G)	Family and friends.	Log book. Applied Tension Release (quick version). Assertiveness training.	
6	6 (G)	Values in life? Finding a balance. Putting it into practice.	Worksheet: values and goals in life.	
7	7 (I)	Goal-setting and plan.	Individual plan.	
8	8 (G)	Maintaining behavior modification. How to handle relapses.	Individual plan.	*
Maintenance program				
10	9 (T)	Follow-up of plan. What works? What does not work? Problem- solving as required.	During the maintenance program, tools and homework are individually adapted.	
12	10 (I)	Follow-up of plan. What works? What does not work? Problem- solving and replanning discussions, as required.		
12	11 (G)	General follow-up of group members. Opportunity to share experience of difficulties and success.		○
16	12 (T)	Same as session 9.		*
20	13 (T)	Same as session 9.		
24	14 (I)	Same as session 10.		Δ
24	15 (G)	Same as session 11.		
32	16 (T)	Same as session 9.		*
40				*
42	17 (T)	Same as session 9.		
48				Δ

G, Group session, two hours; I, Individual session, one hour; T, Telephone contact with the diabetes specialist nurse or the psychologist, 15–20 min; Δ = HbA_{1c}, SDSCA, Swe-PAID-20, HFS, W-BQ, PSS and HAD; ○ = SDSCA; '*' = HbA_{1c}.

with Cronbach's alpha, for the subscale "General diet" was 0.86 and 0.80 for "Blood sugar testing"and "Exercise" respectively. Since the "Specific diet" subscale has shown low inter-item correlation, the two individual items were analyzed separately, as recommended [38].

2.4.3. Diabetes-related emotional effects

Diabetes-related emotional distress was assessed using the Swedish version of the 20-item Problem Areas In Diabetes Scale (Swe-PAID-20) [39]. The questionnaire captures the patient perspective on current emotional burden of diabetes and its treatment. On a 5-point Likert scale, patients rate the degree to which each item is currently a problem for them, from 0 "not a problem" to 4 "a serious problem". The scale produces a total score ranging from 0 to 100, where a higher score indicates greater emotional distress. Cronbach's alpha in the present study was 0.93.

The degree of fear of hypoglycemia was assessed with the 23-item Hypoglycemic Fear Survey (HFS) [40]. The scale comprises the subscales "avoidance behavior" (10 items) and "fear of hypoglycemia" (13 items). Items are rated on a 5-point Likert scale ranging from 0 "never" to 4 "always", giving a sum score ranging from 0 to 92, where a higher score indicates greater fear of hypoglycemia. Cronbach's alpha for the two subscales in the present study was 0.70 and 0.92 respectively, and for the total scale 0.90.

2.4.4. General emotional effects

Well-being was measured in four dimensions using the Well-Being Questionnaire (W-BQ12) [41,42]. The instrument comprises three four-item subscales: "negative well-being" (questions 1–4), with all four items negatively worded; "energy" (questions 5–8), which consists of two positively and two negatively worded items; and "positive well-being" (questions 9–12) with four positively worded items. An overall scale "general well-being", is based on the three subscales, giving a total score of 0–36. The total score for each subscale is 0 to 12. Cronbach's alpha for the different domains varied between 0.77–0.86, and was 0.91 for the total scale.

Mental stress was assessed with the Swedish version of the Perceived Stress Scale (PSS) [43], which comprises 14 items measuring the degree to which situations in one's life are appraised as stressful. On a 5-point Likert scale, respondents rate the degree from 0 "never" to 5 "very often" according to these different situations. Cronbach's alpha in the current study was 0.90.

Anxiety and depression were assessed with the Swedish Hospital Anxiety and Depression (HAD) scale [44], which consists of 14 items measuring level of anxiety and depression in two separate subscales, divided into 7 items respectively. Each item is answered on a 4-point Likert scale, ranging from 0–3, yielding a Cronbach's alpha of 0.87 and 0.89 in the two subscales respectively.

Assessment of secondary variables took place at baseline, week 12 (SDSCA only), week 24 and week 48 (Table 1).

2.4.5. Demographic and clinical variables

In addition, sociodemographic and clinical data were extracted from the patients' files, including age, sex, diabetes duration, cohabitation, education, paid job, HbA_{1c}, BMI, total amount of insulin per kg, diabetic complications and insulin therapy.

2.5. Statistical analyses

Statistical analyses were performed using the SPSS 14.0 for the Social Sciences (SPSS Inc., Chicago, IL, USA). Descriptive statistics

were used for demographic and clinical characteristics of the study sample. Baseline differences between groups were assessed by an unpaired *t* test. For categorical variables, chi-square tests were used. Analyses were carried out on an intention-to-treat basis, which covered all the patients who actually entered the intervention ($n = 40$) and control group ($n = 39$) after randomization and presented values at a given follow-up point. Thus, the intention-to-treat analyses included five patients, of whom four originated from the intervention group and one was from the control group. These patients withdrew from the study for the following reasons; "moved out of area" ($n = 1$), "deceased" ($n = 1$), "pregnant" ($n = 2$) and "diagnosed with cancer, did not want to continue" ($n = 1$). Thus, there were 69 patients who completed the intervention, whereas the final analyses involved 74 patients.

2.5.1. Primary variable

Baseline value was the computed mean of the two HbA_{1c}s during the last year before intervention and the value measured when entering the study. A multivariate analysis of covariance (MANCOVA) was performed, where the results of HbA_{1c} (weeks 8 to 48) were the dependent variables, and 'group' was the independent variable. Results of HbA_{1c} at baseline, sex, BMI and duration of diabetes were added as covariates in the model. The MANCOVA was performed on imputed data, where data for weeks 16, 24, 32, 40 and 48 were imputed for one subject; at weeks 32, 40, and 48 for two subjects; and at week 48 for three subjects, using the expectation-maximization (EM) algorithm in the SPSS missing values module. In the case of a significant overall group effect, post hoc analyses were performed each week, using analysis of covariance (ANCOVA), and

Table 2
Baseline characteristics of the study participants ($n = 74$).

	All patients	Intervention group	Control group	*P*-value
n	74	36	38	
Age (years)	41.2 (12.3; 19–65)	41.1 (11.7; 23–65)	41.4 (12.9; 19–64)	0.914
Sex (male/female) (% female)	36/38 (51.4)	20/16 (44.4)	16/22 (57.9)	0.247
Diabetes duration (years)	21.6 (10.8; 5–48)	19.9 (9.4; 5–44)	23.2 (11.8; 5–48)	0.179
Cohabitation, *n* (%)				
Living alone	21 (28.4)	12 (33.3)	9 (23.7)	0.510
Living with a partner/other	53 (71.7)	24 (66.7)	29 (76.3)	0.510
Education, *n* (%)				
Swedish 'grundskolan' (grades 1–9)	8 (10.8)	2 (5.6)	6 (15.8)	0.156
Upper secondary school	35 (47.3)	18 (50.0)	17 (44.7)	0.650
University	31 (41.9)	16 (44.4)	15 (39.5)	0.665
Paid job, *n* (%)	53 (71.6)	27 (75.0)	26 (68.5)	0.530
Clinical parameters				
$HbA1_C$ (%)[a]	8.5 (0.8; 7.1–11.4)	8.5 (0.9; 7.5–11.4)	8.5 (0.8; 7.1–11.2)	0.989
BMI (kg/m^2)[b]	24.8 (3.0; 17.9–29.9)	24.8 (3.3; 17.9–29.9)	24.8 (2.7; 20.1–29.6)	0.967
Total amount of insulin per kg (U)	0.7 (0.2; 0.4–1.2)	0.7 (0.2; 0.4–1.2)	0.7 (0.4–1.2)	0.650
Diabetic complications, *n* (%)				
Retinopathy	42 (56.8)	20 (55.6)	22 (57.9)	0.839
Peripheral neuropathy	11 (14.9)	4 (11.1)	7 (18.4)	0.377
Overt nephropathy[c]	5 (6.8)	1 (2.8)	4 (10.5)	0.184
Coronary heart disease	3 (4.1)	0 (0)	3 (7.9)	0.085
Cerebrovascular disease	1 (1.4)	0 (0)	1 (2.6)	0.327
Insulin therapy, *n* (%)				
MDI[d] (>3 injections/day)	63 (85.1)	31 (86.1)	32 (84.2)	0.818
CSII[e]	11 (14.9)	5 (13.9)	6 (15.8)	0.818
SDCSA				
General diet	4.8 (1.6)	4.6 (1.7)	5.1 (1.4)	0.313
Specific diet	4.0 (1.8)	3.9 (1.7)	4.0 (1.9)	0.806
Exercise	2.3 (2.0)	2.5 (2.1)	2.0 (1.8)	0.282
Blood sugar testing	3.0 (2.7)	2.9 (2.7)	2.9 (2.8)	0.956
Medication	5.3 (3.1)	4.8 (3.3)	5.8 (2.9)	0.070
Hypoglycemia	5.3 (4.0)	5.2 (4.1)	5.4 (4.0)	0.702
PAID	32.3 (18.8)	31.1 (20.4)	33.4 (17.3)	0.465
HFS				
Behavior avoidance	17.3 (6.1)	16.7 (6.9)	17.8 (5.3)	0.338
Worry or fear	12.9 (9.5)	11.6 (8.9)	14.2 (10.1)	0.349
Total	30.2 (13.9)	28.3 (14.4)	32.0 (13.3)	0.211
W-BQ				
Negative well-being	2.7 (2.5)	2.6 (2.6)	2.8 (2.5)	0.673
Energy ($n = 73$)	5.6 (3.1)	5.3 (3.1)	5.8 (3.1)	0.590
Positive well-being	6.9 (2.8)	6.8 (2.9)	7.0 (2.7)	0.586
Total ($n = 73$)	21.7 (7.4)	21.5 (7.4)	22.0 (7.5)	0.851
PSS	24.0 (9.5)	24.3 (10.3)	23.7 (8.8)	0.916
HAD				
Anxiety	6.5 (4.8)	6.7 (5.1)	6.3 (4.6)	0.871
Depression	4.4 (4.0)	4.5 (3.7)	4.3 (4.2)	0.459

Data are means (SD; range) or number of cases (%).
[a] Reference value < 5.2%.
[b] Reference value 19–24 kg/m^2 (women), 20–25 kg/m^2 (men).
[c] Albumin excretion > 200 μg/min.
[d] Multiple daily injections.
[e] Continuous subcutaneous insulin infusion (insulin-pump therapy).

using the same covariates as in the MANCOVA. To ensure that the assumptions of the tests were fulfilled, standard analyses of residuals were performed for the ANCOVA models each week. In the case of apparent non-normality, data were log-transformed, and in the case of extreme outliers, the analyses were performed both with and without the outliers as a consistency check.

2.5.2. Secondary variables

An ANCOVA was performed at weeks 12, 24 and 48, using the same covariates as in the primary analysis. Missing values were imputed using the same imputation method as for the primary variable. Model checks have been carried out and results are presented in the same way as for the primary variable.

3. Results

Participants' baseline characteristics are given in Table 2, which demonstrates no significant differences between the two groups. Furthermore, there were no significant differences regarding the same baseline characteristics, between patients from the two hospitals, nor between those who chose to participate in the study and those who declined.

Participants (n = 36) attended an average of 6.8 + 1.2 (range 4–8) of the eight basic sessions in the CBT-based intervention, and the attendance rate at follow-up sessions was 3.6 + 0.6 (range 2–4) out of four sessions.

3.1. Primary variable

The results of the final analyses are presented in Table 3. At 8 weeks and further on up to 48 weeks, a significant difference was shown between the intervention group and the control group. The difference appeared to be greatest at 24 weeks. One outlier was identified in the control group, pointing in a positive direction. When excluding this outlier from the analyses, results tended to be more positive but without a determining effect on the significance level. Therefore all analyses are based on all cases.

Table 3
HbA_{1c} levels (%) at weeks 8–48.

Time point (week)	Intervention (n = 36)	Control (n = 38)	Difference (95% CI)	P-value
8	7.58	8.25	−0.67 (−0.97 to −0.36)	<0.001***
16	7.47	8.36	−0.89 (−1.30 to −0.48)	<0.001***
24	7.50	8.44	−0.94 (−1.36 to −0.51)	<0.001***
32	7.68	8.40	−0.72 (−1.13 to −0.31)	0.001**
40	7.81	8.37	−0.56 (−0.95 to −0.16)	0.007**
48	7.72	8.21	−0.49 (−0.87 to −0.11)	0.012*

Data are adjusted means.
* <0.05.
** <0.01.
*** <0.001.

3.2. Secondary variables

After 12 weeks (data not included in Table 4) significant differences were observed regarding SMBG, which was improved with a mean value of 5.17 in the intervention group vs. 3.31, difference 1.87 (0.77 to 2.96), $P < 0.01$. In Table 4, results of secondary variables at the 24- and 48-week follow-up are presented. After 24 weeks, significant differences were found between groups regarding SMBG, incidence of hypoglycemia,

Table 4
Effects of self-care behaviors, diabetes-related emotional and general emotional aspects at 6-month and 1-year follow-up in the intervention and the control group.

	At 24-week follow-up				At 48-week follow-up			
	Intervention (n = 36)	Control (n = 38)	Difference (95% CI)	P-value	Intervention (n = 36)	Control (n = 38)	Difference (95% CI)	P-value
Effects of self care								
SDSCA								
General diet	5.12	4.77	0.36 (−0.15 to 0.87)	0.161 ns	5.12	4.95	0.18 (−0.30–0.65)	0.458 ns
Specific diet item 1[a]	3.12	2.65	0.47 (−0.46 to 1.40)	0.316 ns	4.04	3.58	0.46 (−0.58–1.51)	0.38 ns
Specific diet item 2[b]	4.91	4.29	0.62 (−0.001 to 1.24)	0.05 ns	4.93	4.57	0.35 (−0.20–0.90)	0.20 ns
Exercise	2.62	2.18	0.44 (−0.37 to 1.25)	0.284 ns	2.62	2.31	0.31 (−0.48–1.11)	0.436 ns
Blood sugar testing	5.34	3.14	2.20 (1.19 to 3.21)	<0.001***	4.74	3.35	1.39 (0.35–2.44)	0.010*
Medication	6.53	6.05	0.49 (−0.25 to 1.22)	0.191 ns	6.05	6.27	−0.22 (−1.04–0.60)	0.590 ns
Hypoglycemia	7.14	4.81	2.33 (0.46 to 4.21)	0.016*	6.77	4.43	2.34 (0.01–4.66)	0.049*
Diabetes-related emotional effects								
Swe-PAID-20	24.43	30.60	−6.18 (−11.30 to −1.05)	0.019*	22.92	29.80	−6.88 (−11.50–−2.25)	0.004**
HFS								
Beh avoid	14.88	16.36	−1.49 (−3.15 to 0.19)	0.081 ns	15.30	17.94	−2.64 (−4.51–−0.76)	0.006**
Worry/Fear	11.50	12.73	−1.23 (−3.85 to 1.39)	0.351 ns	11.84	12.12	−0.28 (−2.39–1.82)	0.790 ns
Total	26.43	29.04	−2.61 (−6.26 to 1.05)	0.159 ns	27.19	30.02	−2.83 (−6.02–0.36)	0.081 ns
General emotional effects								
W-BQ								
NegW-B	2.13	2.93	−0.80 (−1.55 to -0.05)	0.036*	1.84	2.80	−0.96 (−1.69–−0.22)	0.011*
Energy	6.02	5.99	0.02 (−0.89 to 0.94)	0.960 ns	6.95	6.14	0.81 (−0.22–1.84)	0.121 ns
PosW-B	6.98	6.82	0.16 (−0.76 to 1.08)	0.729 ns	7.43	6.85	0.59 (−0.27–1.44)	0.178 ns
Total	22.86	21.88	0.99 (−0.94 to 2.93)	0.310 ns	24.57	22.18	2.39 (0.30–4.49)	0.026*
PSS	21.12	23.61	−2.48 (−5.54 to 0.57)	0.110 ns	20.01	23.74	−3.73 (−6.73–−0.72)	0.016*
HAD								
Anxiety	5.39	6.51	−1.12 (−2.39 to 0.15)	0.082 ns	5.07	6.32	−1.25 (−2.49–−0.01)	0.048*
Depression	3.76	4.58	−0.81 (−2.25 to 0.62)	0.262 ns	3.51	5.09	−1.59 (−2.98–−0.18)	0.027*

Data are adjusted means. Ns, not significant.
[a] Item's wording: On how many of the last seven days did you eat five or more servings of fruits and vegetables?
[b] Item's wording: On how many of the seven days did you eat high fat foods?
* <0.05.
** <0.01.
*** <0.001.

diabetes-related distress and negative well-being. At 48 weeks, significant differences were maintained regarding the same factors. In addition, significant changes were shown in general well-being, perceived stress, anxiety, depression and avoidance of hypoglycemia, which all improved within the intervention group. During the intervention no patient reported an episode of severe hypoglycemia, requiring hospital admission. One episode of ketoacidosis requiring hospital admission occurred in the intervention group.

4. Discussion and conclusion

4.1. Discussion

Significant differences in HbA_{1c} between the two groups were observed early in the study at week 8, and were maintained throughout the study up to week 48. To our knowledge, previous behavior-oriented investigations of this targeted group of patients have not provided such significant and long-lasting effects as those generated here. Unlike our study sample, participants in the study of van der Ven et al. [23] displayed a less favorable psychological profile, with 45% likelihood of suffering from clinical depression, which might explain that their results differed from ours. However, it also seems meaningful to speculate upon the possible importance of the components added to the present intervention. Our intervention program included formats that were both group- and individually oriented, which allowed advantages of group processes to be combined with elements of necessary individualization. This combination might have facilitated motivational aspects of the program. Moreover, concrete, everyday-life behaviors and especially self-care behaviors were central and explicit targets in the program, both in the themes and behavioral tools introduced during the sessions, through the use of the "biofeedback" methodology, and through the weekly and systematic follow-up of homework assignments, which included actual self-care behaviors. In addition, the rather intensive and long-term maintenance program might also have been of value in order to help maintain the behavior changes achieved, as well as to tackle actual or potential relapses. An important area for future research is to evaluate outcomes when using a no-maintenance group, for comparison with the current research design.

Improvements in HbA_{1c} should always raise concerns of hypoglycemia [3]. Increased incidence of hypoglycemia was observed in our intervention group, with significant between-group differences at weeks 24 and 48. This must be acknowledged as an important possible negative effect of the CBT program that needs to be weighed up against its benefits, and/or closely watched for, when this program is implemented in the future. With respect to hypoglycemia, it is notable that the participants had altered their avoidance behavior, but not their fear of hypoglycemia at the end of the study. We find this interesting, given that our patients rated avoidance of hypoglycemia higher than fear of hypoglycemia.

Our study cannot provide any answers to the question of what ingredients were the most important in view of the favorable effects registered. Apart from conceivable key elements such as the group/individualized and maintenance formats, it is reasonable to assume that more frequent SMBG is important. Increased frequency of SMBG was observed at 12-week follow-up and continued throughout the study. The early effect on this behavior change is not surprising, given that SMBG was part of the participants' daily homework during the first 8 weeks. It is interesting, though, that the behavior was maintained on a more self-chosen basis among participants, indicating the possibility of a newly learnt behavior of self-care, which is an important outcome in itself. A recent multicenter analysis [45] indicated that more frequent SMBG is associated with better metabolic control in patients with type 1 diabetes. However, another study from a Swedish university hospital [46] reported that only 49% of type 1 diabetes patients performed SMBG every day. It should be noted that SMBG is fully reimbursed in Sweden, so cost is no barrier to SMBG here. It is reasonable to assume that raised awareness of blood glucose levels may lead to patients taking action in either direction, e.g. modifying current insulin therapy. To ascertain whether such actions were taken, van der Ven et al. [23] asked about adjustment of insulin, while we asked about adherence regarding the same factor. This might have led to respondents giving different answers, since we found no significant differences in this self-care behavior.

Participants in our intervention group were exposed to increased attention for almost one year. We cannot rule out the possible effect of this alone. In long-term follow-up, it has been shown that the amount of contact time between care providers and patients is a significant predictor of effect on glycemic control [47]. In the present study, contact time between care providers and patients was about 23 h during one year. The control group was also paid a certain amount of attention, in that they were sent results on CGMS and HbA_{1c}-levels, taken regularly, by mail.

In this intervention, participants were asked to deal with attitudes towards diabetes and self-care. As part of their homework they were instructed to observe negative feelings and to reflect upon values in life. Dealing with these aspects could have made the participants more overwhelmed and worried about life and diabetes. We found it reassuring to note that after completing the program our participants showed only preserved or improved scores on all psychological variables.

Regarding its clinical implications, our study must be interpreted with caution. Firstly, the results have proved to be statistically and clinically promising in a selected group of patients, where about 32% of the 230 patients who were eligible, have been analyzed. The fact that the others declined to participate or withdrew from the study, calls for some concern on the issue of external validity. It seems reasonable to believe that this kind of intervention may work very well for those who are more motivated, or for whom there are fewer barriers to attending sessions like these. Similar to previous studies, attrition tended to dominate at an early stage before intervention and then decreased as time progressed [48,49]. Further research is needed to understand why so many eligible individuals refused to participate. Different approaches are probably needed for this group of patients. Secondly, the participants in our study improved their glycemic control to a level that is far from satisfactory, with an HbA_{1c} level of 7.7% at 48 weeks. Nevertheless, we believe that this can still be considered a positive result, given the fact that behavior modification involves different stages of change [50] and usually is a time-consuming process. Thirdly, a formal analysis of the secondary outcomes as mediators of the effect in the intervention on glycemic control would have increased the significance of this study [51]. For future research such an evaluation will require a larger sample size than ours.

4.2. Conclusion

In summary, we suggest that this CBT-based intervention is a promising approach in improving outcome in poorly controlled adult type 1 diabetes patients. We believe that the reduction in HbA_{1c} obtained here, should be regarded as clinically relevant in light of available DCCT data [3,52]. The approach is worthy of further evaluation in research and clinical practice. We also suggest further research on type 2 diabetes and on targeted groups

of diabetes patients, e.g. those who have to deal with eating disorders, fear/avoidance of hypoglycemia, depression, anxiety and stress. Research in this area of problematic diabetes is still scarce [53] and a CBT-based approach might be beneficial.

4.3. Practice implications

Given the results above, diabetes care may benefit from a CBT-based approach delivered both in group and individual formats, including tools commonly used in CBT and incorporating a structured maintenance program. To evaluate this behavior-oriented approach in clinical practice, there is a need for specially educated diabetes care teams, trained in the current approach, as well as cooperation between diabetes care teams and psychologists trained in CBT.

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